

Do Attention-Derived Hopfield Energies Contain Hallucination Signal?

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Study project

Problem

- Hallucination detectors usually inspect generated text, external evidence, or model confidence.
- I ask a diagnostic question: can hallucination signal be seen inside the model's attention geometry?
- Focus: frozen language models, no finetuning of the base model.

Main claim

Q/K energy and attention-derived summaries contain transferable factuality signal, but scalar energy is not a universal hallucination score.

- Transformer attention compares queries and keys.
- Modern Hopfield networks also perform associative retrieval through softmax similarity.
- If a response is weakly supported, its retrieval geometry may look different.

$$E_t = -\sqrt{d_h} \log \sum_{i \leq t} \exp \left(\frac{q_t^\top k_i}{\sqrt{d_h}} \right)$$

Hopfield-style energy from attention Q/K states

Method

- ① Run frozen LLM on prompt + response.
- ② Extract Q/K tensors after RoPE from selected attention layers.
- ③ Score response tokens only.
- ④ Compute Hopfield energy summaries and prompt-attention summaries.
- ⑤ Train simple logistic-regression probes with 5-fold CV.

Baselines

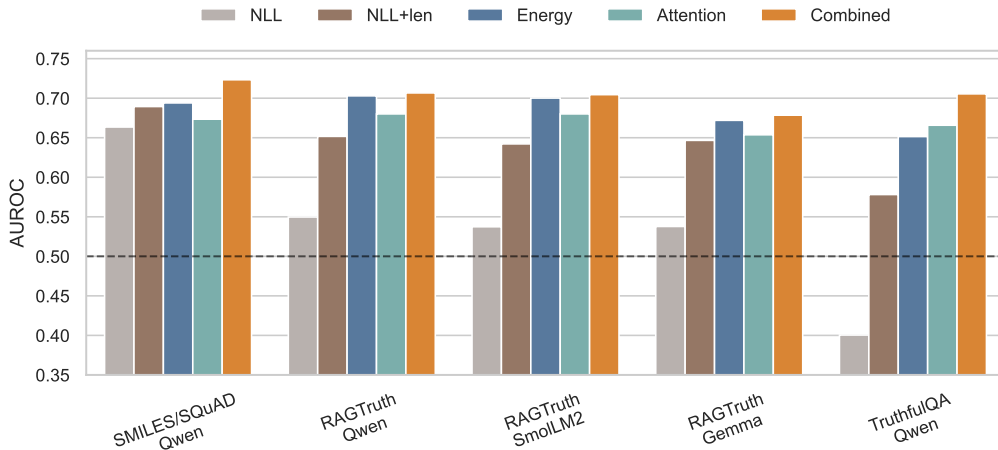
Response length, response negative log-likelihood, and NLL + length.

Datasets and Models

Dataset	Hallucination type	Models
SMILES/SQuAD	context-grounded QA	Qwen2.5-0.5B
RAGTruth	context-grounded RAG	Qwen2.5-0.5B, SmolLM2-360M, Gemma-3-1B
TruthfulQA	open-domain false answers	Qwen2.5-0.5B

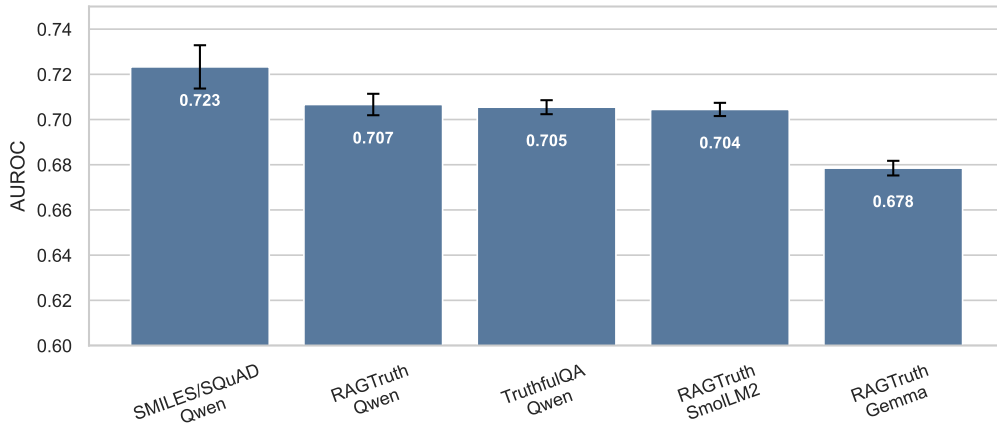
- Learned probes are repeated over CV seeds: 13, 21, 42, 87, 101.
- AUROC is the main metric because hard-label F1 depends strongly on threshold calibration.

Main Result: Feature Families



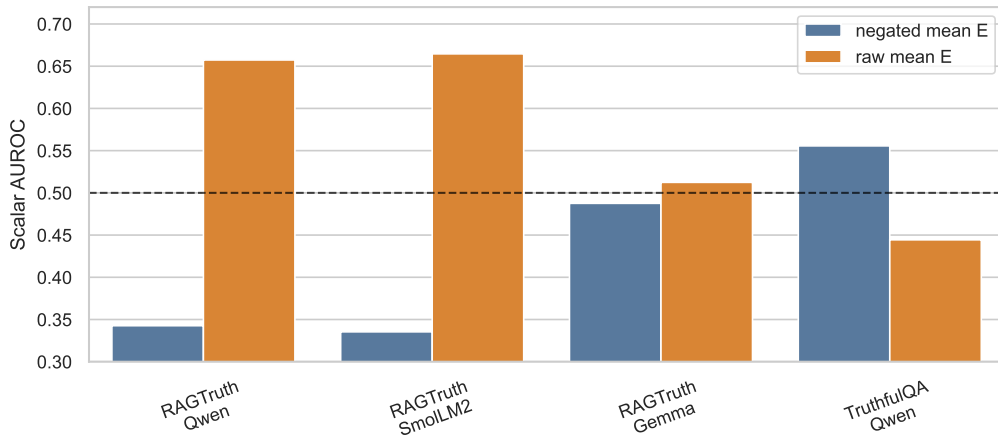
Combined Q/K-derived probes improve over response likelihood baselines.

Transfer Across Models



Qwen and SmolLM2 transfer strongly on RAGTruth; Gemma is weaker but remains above NLL/length.

Important Caveat: Scalar Energy Direction



Scalar energy is not a universal monotonic hallucination score.

Setting	Best AUROC mean $\pm$ std
SMILES/SQuAD + Qwen2.5-0.5B	0.723 $\pm$ 0.010
RAGTruth + Qwen2.5-0.5B	0.707 $\pm$ 0.005
RAGTruth + SmolLM2-360M	0.705 $\pm$ 0.003
RAGTruth + Gemma-3-1B	0.679 $\pm$ 0.003
TruthfulQA + Qwen2.5-0.5B	0.706 $\pm$ 0.003

- Stable across CV seeds.
- Strongest evidence: RAGTruth transfer from Qwen to SmolLM2.
- Weakest transfer: Gemma, suggesting architecture/layer sensitivity.

Conclusion

What worked

Attention-derived Q/K energy and prompt-attention summaries contain hallucination/factuality signal beyond response likelihood.

What did not become a universal rule

A single scalar rule like “higher energy means hallucination” is too strong.

Next steps

Per-head analysis, layer sensitivity, larger models, and cleaner out-of-domain transfer tests.

Code, report, and reproduction
commands

`https://github.com/eivankin/
hopfield-energy-hallucination-detection`

- feature extraction
- public dataset conversion
- cached multi-seed evaluation
- final report and figures

